

Kilian Shi

Summer 2027 AI / ML Internship | Available May 12 – Aug 17

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EDUCATION

Carnegie Mellon University

May 2028

Master of Science in Mobile and IoT Engineering

Chongqing University

Jul 2026

Bachelor of Engineering in Computer Science and Technology

GPA: 88/100

PROFESSIONAL EXPERIENCE

Ericsson (China)

Sep 2025 – Nov 2025

AI Engineering Intern

- Enhanced a Django RAG platform with BM25, ChromaDB, and RRF; diagnosed a ranking bug returning the correct answer second.
- Built LLM-generated QA and LLM-as-a-judge evaluation with hit rate and MRR to guide retrieval tuning.
- Separated retrieval and reranking inputs; added HyDE query rewriting for forum search.
- Added DB refresh, upload isolation, and HITL review tools; briefed a senior manager in Ericsson Sweden on the full project.

Beijing Silu Innovation Technology

Jun 2025 – Jul 2025

Back-End Development Intern

- Migrated enterprise web modules from Spring MVC to Spring Boot and Vue.
- Automated code-repository documentation through a reusable web scraper.

RESEARCH EXPERIENCE

University of Notre Dame

Jul 2025 – Present

Research Intern, iSURE

SIGN: Shape-Aware Sperm Head Morphology Classification

Aug 2025 – Present

Ongoing Research

- Proposed INR shape priors for a frozen SAM3 classifier; reached 69.1% macro-F1 on SCIAN-PA and 98.7% on HuSHeM.
- Encoded masks into 256-D SIREN/DeepSDF latents and fused shape priors through FiLM and contrastive training.

D-GEM: Sparsely Supervised Surgical Video Segmentation

Jul 2025 – Feb 2026

MICCAI 2026

- Co-designed DINOv3 dual-memory gating and cosine-similarity grouping based on SAM2/SAM3 propagation failures.
- Method reached 74.2% mean mIoU on 3 benchmarks with 10 labels/video, 10.3 points above re-prompted SAM3.

UKAST: KAN-Enhanced Swin Transformer for Medical Segmentation

Aug 2025 – Oct 2025

IEEE ISBI 2026 | arXiv:2511.04084

- Refined GR-KAN integration into Swin MLP blocks and implemented model components with coauthors.
- Ran most ablations in shared W&B experiments; gained 0.4 (2D) and 1.3 (3D) mean Dice points over SwinUNETR-RC.
- Led weekly KAN paper reviews and architecture discussions, turning findings into experiments and manuscript updates.

Chongqing University

Sep 2025 – May 2026

Undergraduate Researcher

Dental Lesion Recognition

Undergraduate Thesis

- Fine-tuned YOLOv8-Seg on 3,616 tooth instances, attaining 0.782 mask mAP@0.5.
- Combined context, FiLM, and Rank-MIL in Swin; reached 0.861 macro-F1 and 110 FPS (RTX A4000, batch 1).
- Built a web prototype with per-tooth scores and overlays; reported weekly experiments to my faculty advisor.

SKILLS

Languages: Python, C/C++, Java, SQL

ML / AI: PyTorch, JAX, scikit-learn, Hugging Face, Computer Vision, LLMs, RAG, Weights & Biases

Engineering: Git, Docker, Django, FastAPI, Spring Boot, Vue, ChromaDB, GCP

Data Systems: Spark, Kafka, Neo4j